

# NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

## 0. Motivasyon ve Yontem

Mevcut missing stratejisi M3 (>%50 NaN icin is\_missing flag + medyan impute) NB14 panel-transfer ablasyonuyla secildi, ama eksiklik-etiket korelasyonu sistematik olculmedi.

Bu notebook: her panelde -- sabit ve ozdes sutunlar drop edildikten sonra -- eksiklik>0 olan her sutun icin is\_missing flag uretir, flag ile Label arasindaki phi (phi) katsayisini ve sinif-bazli eksiklik oranlarini hesaplar. Sutunlar miss\_ratio>0.50 (G\_HIGH) ve miss\_ratio<=0.50 (G\_LOW) olarak iki gruba ayrilir; her grup icin mean\_phi (isaretli), mean\_abs\_phi (toplam sinyal) ve fluctuation=std(phi) (grup-ici tutarlilik) raporlanir.

Bu bir olcum, tahmin degildir: train/test split yok, impute yok. Flag'ler temiz+HAM eksiklikten uretildi (impute ONCESI).

## 0b. On Temizlik: Sabit ve Ozdes Sutunlar

CLAUDE.md Sozlesmeler madde 3-4 geregi diger notebooklarda (NB15, NB36-39) uygulanan temizlik burada da uygulandi: sabit sutunlar (nunique<=1) ve ozdes sutun ciftlerinden biri (583 cift, cogu AL bloklarinda) drop edildi. Nedeni: ozdes sutunlar (orn. MASTER'da AL\_1..AL\_6) ayni missing-flag sinyalinin birebir kopyasidir -- temizlik olmadan 'en yuksek phi' listeleri tek bir sinyalin N kez tekrari gibi gorunur, bagimsiz kanit sayisi abartilmis olur.

| Panel  | Ham Sutun | Sabit | Ozdes Cift | Drop Toplam | Temiz Sutun |
|--------|-----------|-------|------------|-------------|-------------|
| MASTER | 351       | 57    | 583        | 63          | 288         |
| KANSER | 351       | 69    | 694        | 71          | 280         |
| PAH    | 351       | 91    | 886        | 93          | 258         |
| CFTR   | 351       | 70    | 696        | 72          | 279         |

## Karar Kurali

| Bulgu                             | Anlam                          | Aksiyon                             |
|-----------------------------------|--------------------------------|-------------------------------------|
| G_LOW mean phi  yuksek (>-0.15)   | M3 degerli flag'leri kaciriyor | Esigi dusur / secici flag seti      |
| G_HIGH yuksek + dusuk fluctuation | Eksiklik guclu, tutarli sinyal | M3 dogru; blogu modelde tut         |
| Bir grupta yuksek fluctuation     | Sinyal birkac sutunda          | O sutunlari izole et, kalani drop   |
| AL_16..25 (KANSER) yuksek phi     | Bilinen leakage riski teyidi   | Ikı model (ile/olmadan) karsilastir |

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### 1. Ozet: 4 Panel x 2 Grup

| Panel  | Grup   | n_cols | mean_phi | mean_abs_phi | fluctuation | mean_delta |
|--------|--------|--------|----------|--------------|-------------|------------|
| MASTER | G_HIGH | 140    | +0.2357  | 0.2357       | 0.0459      | +0.2367    |
| MASTER | G_LOW  | 148    | +0.1188  | 0.1363       | 0.0854      | +0.1369    |
| KANSER | G_HIGH | 191    | +0.4164  | 0.4164       | 0.0608      | +0.4099    |
| KANSER | G_LOW  | 89     | +0.1943  | 0.2531       | 0.2234      | +0.2252    |
| PAH    | G_HIGH | 150    | -0.0414  | 0.0474       | 0.0307      | -0.0562    |
| PAH    | G_LOW  | 108    | +0.0515  | 0.0844       | 0.1091      | +0.0777    |
| CFTR   | G_HIGH | 31     | +0.2863  | 0.2863       | 0.1540      | +0.2684    |
| CFTR   | G_LOW  | 237    | +0.0022  | 0.0752       | 0.0938      | -0.0007    |

Saglama noktası: KANSER'de AL\_16..25 (AL\_MISSINGNESS\_LEAKAGE\_RISK) blogu, tum sutunlar arasinda en yuksek abs\_phi degerlerini vermeli. Top-10 kesisim: 10/10 sutun -> ['AL\_16', 'AL\_17', 'AL\_18', 'AL\_19', 'AL\_20', 'AL\_21', 'AL\_22', 'AL\_23', 'AL\_24', 'AL\_25']

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### 2. MASTER - G\_HIGH (miss\_ratio > %50)

| col   | miss% | miss_patho% | miss_benign% | delta  | phi    |
|-------|-------|-------------|--------------|--------|--------|
| AL_1  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_5  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_6  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_2  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_4  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_3  | 0.900 | 0.962       | 0.728        | +0.235 | +0.346 |
| AL_27 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_38 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_37 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_35 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_33 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_32 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_31 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_30 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_29 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_28 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_36 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_34 | 0.914 | 0.966       | 0.770        | +0.197 | +0.310 |
| AL_19 | 0.640 | 0.728       | 0.396        | +0.332 | +0.306 |
| AL_23 | 0.640 | 0.728       | 0.396        | +0.332 | +0.306 |
| AL_16 | 0.640 | 0.728       | 0.396        | +0.332 | +0.306 |
| AL_17 | 0.640 | 0.728       | 0.396        | +0.332 | +0.306 |
| AL_18 | 0.640 | 0.728       | 0.396        | +0.332 | +0.306 |
| AL_24 | 0.640 | 0.728       | 0.398        | +0.331 | +0.305 |
| AL_25 | 0.640 | 0.728       | 0.398        | +0.331 | +0.305 |

Ozet -- n\_cols=140 | mean\_phi=+0.2357 | mean\_abs\_phi=0.2357 | fluctuation=0.0459 | top3=AL\_1, AL\_5, AL\_6

### 2. MASTER - G\_LOW (0 < miss\_ratio <= %50)

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| AL_300 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_305 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_315 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_314 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_313 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_311 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_308 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_306 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_304 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_318 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_302 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_301 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_298 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_297 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_296 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_295 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |
| AL_317 | 0.443 | 0.506       | 0.271        | +0.235 | +0.209 |

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|        |       |       |       |        |        |
|--------|-------|-------|-------|--------|--------|
| AL_310 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_319 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_328 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_334 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_321 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_331 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_330 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |
| AL_329 | 0.443 | 0.506 | 0.271 | +0.235 | +0.209 |

Ozet -- n\_cols=148 | mean\_phi=+0.1188 | mean\_abs\_phi=0.1363 | fluctuation=0.0854 | top3=AL\_300, AL\_305, AL\_315

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### 2. KANSER - G\_HIGH (miss\_ratio > %50)

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| AL_20  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_22  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_25  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_24  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_16  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_17  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_18  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_19  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_23  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_21  | 0.686 | 0.862       | 0.292        | +0.570 | +0.568 |
| AL_215 | 0.773 | 0.918       | 0.450        | +0.468 | +0.516 |
| AL_217 | 0.773 | 0.918       | 0.450        | +0.468 | +0.516 |
| AL_214 | 0.773 | 0.918       | 0.450        | +0.468 | +0.516 |
| AL_184 | 0.732 | 0.873       | 0.417        | +0.456 | +0.476 |
| AL_183 | 0.732 | 0.873       | 0.417        | +0.456 | +0.476 |
| AL_2   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_6   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_5   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_4   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_3   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_1   | 0.884 | 0.985       | 0.658        | +0.327 | +0.472 |
| AL_253 | 0.722 | 0.858       | 0.417        | +0.442 | +0.455 |
| AL_250 | 0.722 | 0.858       | 0.417        | +0.442 | +0.455 |
| AL_251 | 0.722 | 0.858       | 0.417        | +0.442 | +0.455 |
| AL_242 | 0.698 | 0.836       | 0.392        | +0.444 | +0.447 |

Ozet -- n\_cols=191 | mean\_phi=+0.4164 | mean\_abs\_phi=0.4164 | fluctuation=0.0608 | top3=AL\_20, AL\_22, AL\_25

### 2. KANSER - G\_LOW (0 < miss\_ratio <= %50)

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| AL_7   | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_12  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_8   | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_14  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_13  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_15  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_11  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_10  | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_9   | 0.446 | 0.604       | 0.092        | +0.513 | +0.477 |
| AL_26  | 0.428 | 0.582       | 0.083        | +0.499 | +0.466 |
| CAT_1  | 0.428 | 0.582       | 0.083        | +0.499 | +0.466 |
| AL_305 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |
| AL_313 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |
| AL_311 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |
| AL_310 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |
| AL_308 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |
| AL_306 | 0.490 | 0.616       | 0.208        | +0.407 | +0.377 |

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|        |       |       |       |        |        |
|--------|-------|-------|-------|--------|--------|
| AL_298 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_304 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_302 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_301 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_300 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_315 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_297 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |
| AL_296 | 0.490 | 0.616 | 0.208 | +0.407 | +0.377 |

Ozet -- n\_cols=89 | mean\_phi=+0.1943 | mean\_abs\_phi=0.2531 | fluctuation=0.2234 | top3=AL\_7, AL\_12, AL\_8

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SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### 2. PAH - G\_HIGH (miss\_ratio > %50)

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| AL_184 | 0.680 | 0.665       | 0.758        | -0.094 | -0.075 |
| AL_183 | 0.680 | 0.665       | 0.758        | -0.094 | -0.075 |
| AL_41  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_57  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_45  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_46  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_47  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_48  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_49  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_50  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_51  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_52  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_53  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_54  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_55  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_56  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_59  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_58  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_43  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_60  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_61  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_62  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_63  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_64  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |
| AL_65  | 0.581 | 0.565       | 0.661        | -0.097 | -0.073 |

Ozet -- n\_cols=150 | mean\_phi=-0.0414 | mean\_abs\_phi=0.0474 | fluctuation=0.0307 | top3=AL\_184, AL\_183, AL\_41

### 2. PAH - G\_LOW (0 < miss\_ratio <= %50)

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| AL_325 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_314 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_317 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_318 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_319 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_322 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_323 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_313 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_326 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_327 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_328 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_329 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_330 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_331 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_333 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_334 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |
| AL_311 | 0.339 | 0.384       | 0.113        | +0.271 | +0.213 |

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|        |       |       |       |        |        |
|--------|-------|-------|-------|--------|--------|
| AL_310 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_306 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_305 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_302 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_301 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_300 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_298 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |
| AL_297 | 0.339 | 0.384 | 0.113 | +0.271 | +0.213 |

Ozet -- n\_cols=108 | mean\_phi=+0.0515 | mean\_abs\_phi=0.0844 | fluctuation=0.1091 | top3=AL\_325, AL\_314, AL\_317



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### 2. CFTR - G\_HIGH (miss\_ratio > %50) [DUSUK-GUVEN: n\_benign=21]

UYARI: CFTR toplam benign sayisi 21. Sinif-bazli eksiklik oranlari (miss\_rate\_benign, delta)

cok gurultulu -- tek basina guvenilmemeli. Bkz. CLAUDE.md CFTR DEGERLENDIRME GUVENI YOK bolumu.

| col   | miss% | miss_patho% | miss_benign% | delta  | phi    |
|-------|-------|-------------|--------------|--------|--------|
| AL_1  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_29 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_27 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_2  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_30 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_31 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_32 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_33 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_34 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_35 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_36 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_37 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_38 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_6  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_5  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_4  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_3  | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_28 | 0.865 | 0.933       | 0.571        | +0.362 | +0.415 |
| AL_25 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_24 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_23 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_22 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_21 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_20 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |
| AL_19 | 0.505 | 0.533       | 0.381        | +0.152 | +0.119 |

Ozet -- n\_cols=31 | mean\_phi=+0.2863 | mean\_abs\_phi=0.2863 | fluctuation=0.1540 | top3=AL\_1, AL\_29, AL\_27

### 2. CFTR - G\_LOW (0 < miss\_ratio <= %50) [DUSUK-GUVEN: n\_benign=21]

| col    | miss% | miss_patho% | miss_benign% | delta  | phi    |
|--------|-------|-------------|--------------|--------|--------|
| EK_3   | 0.441 | 0.500       | 0.190        | +0.310 | +0.244 |
| AL_315 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_296 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_297 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_298 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_300 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_301 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_302 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_304 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_305 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_306 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_308 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_310 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |
| AL_311 | 0.135 | 0.167       | 0.000        | +0.167 | +0.191 |

## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

|        |       |       |       |        |        |
|--------|-------|-------|-------|--------|--------|
| AL_313 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_314 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_317 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_318 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_319 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_321 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_322 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_323 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_325 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_326 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |
| AL_327 | 0.135 | 0.167 | 0.000 | +0.167 | +0.191 |

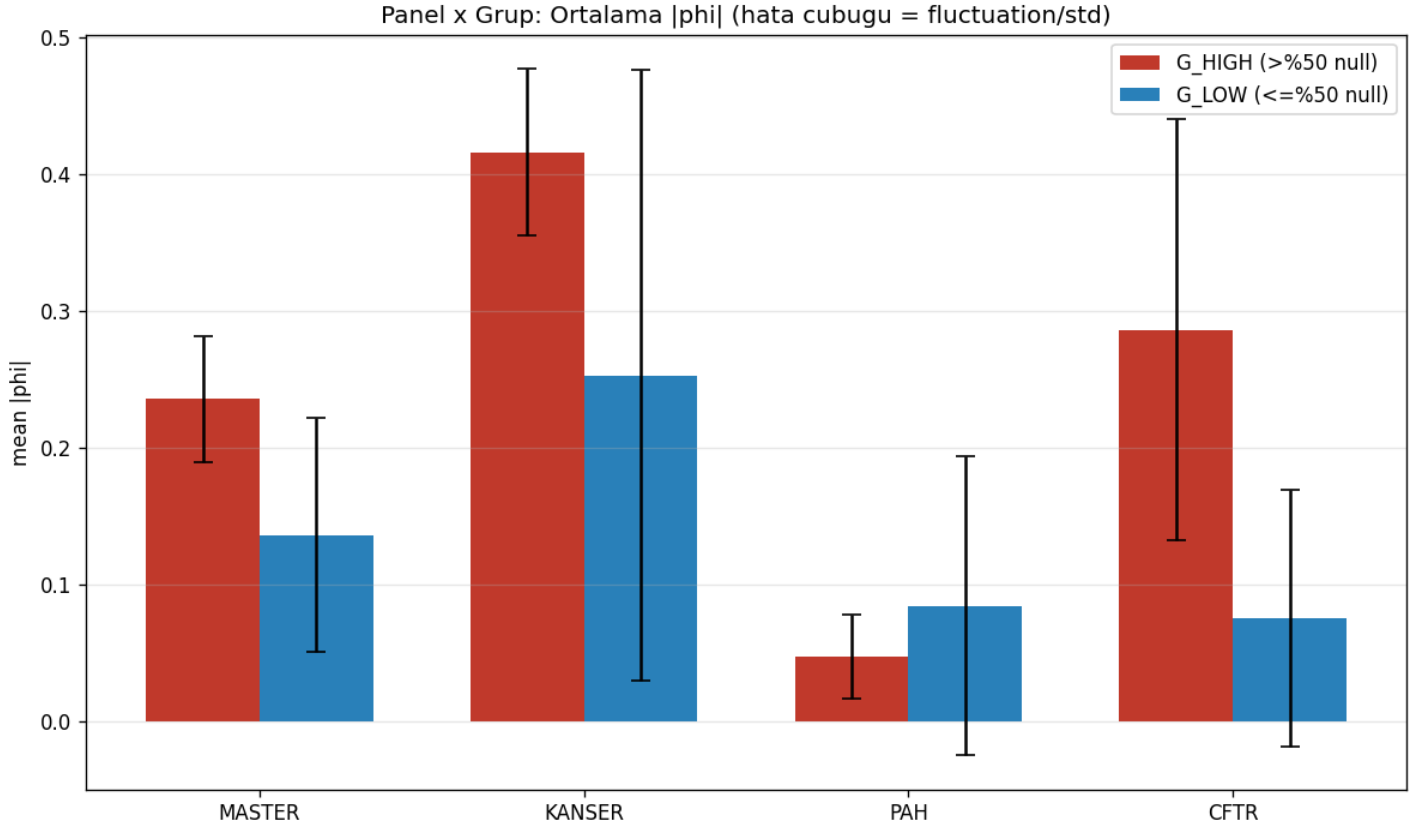
Ozet -- n\_cols=237 | mean\_phi=+0.0022 | mean\_abs\_phi=0.0752 | fluctuation=0.0938 | top3=EK\_3, AL\_315, AL\_296

## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### 3. Gorseller

#### Panel x Grup Ortalama |phi| (fluctuation error-bar)

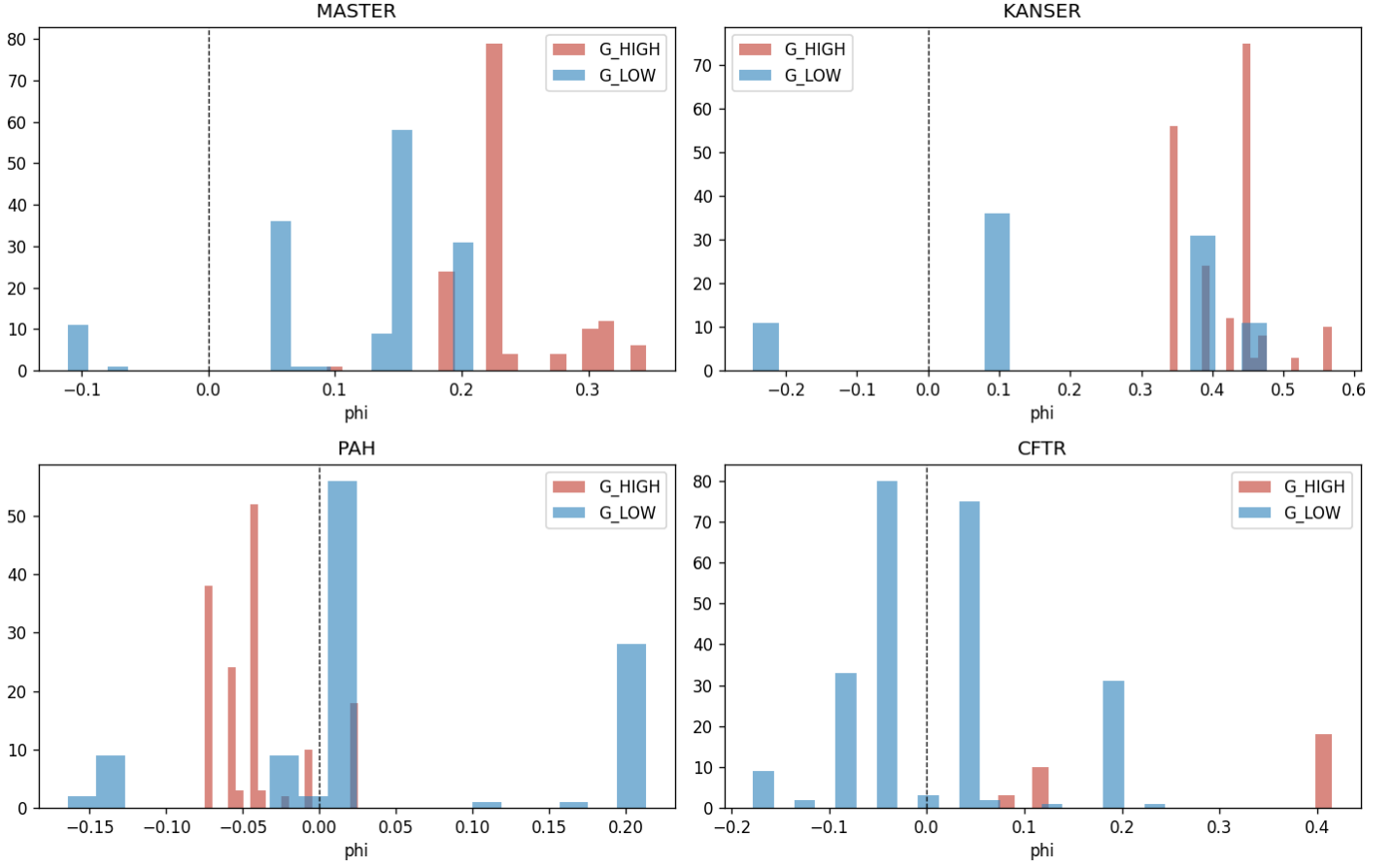


## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### phi Dagilimi: G\_HIGH vs G\_LOW

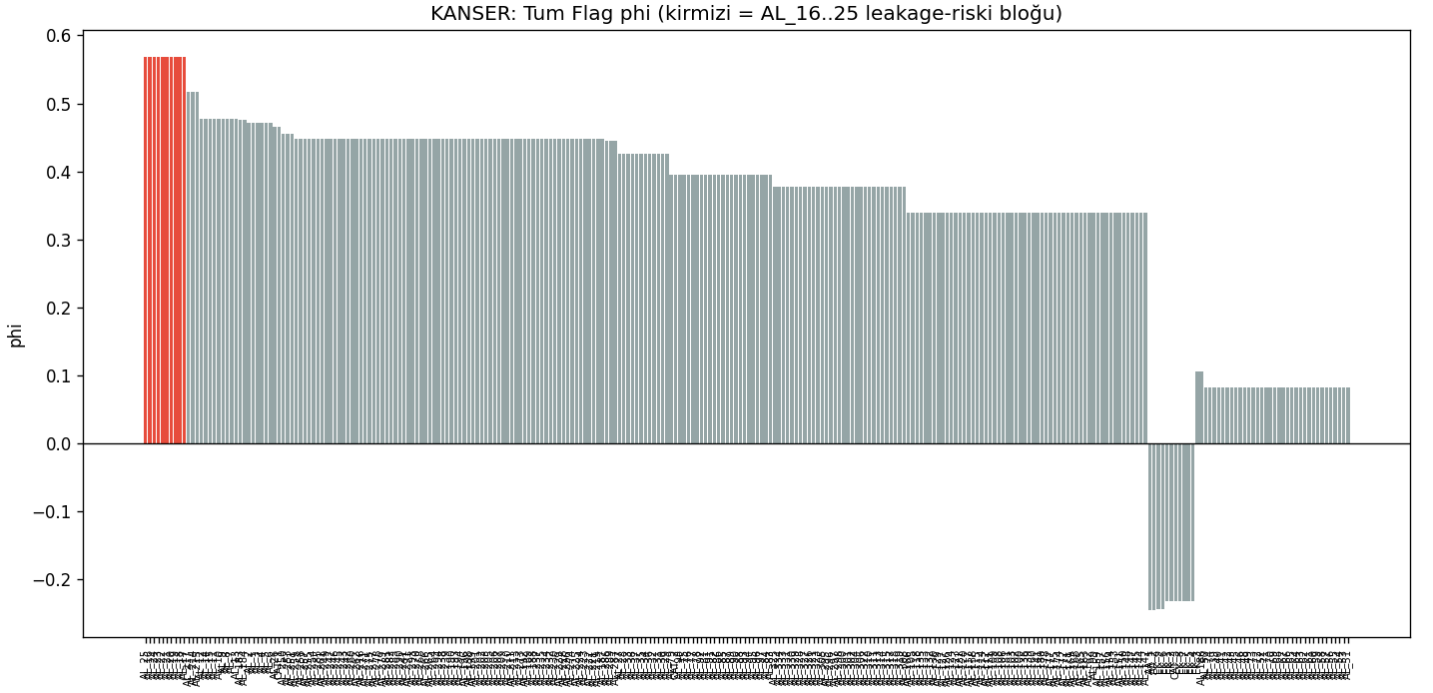
phi Dagilimi: G\_HIGH vs G\_LOW



## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### KANSER AL\_16..25 Leakage Bloğu Vurgulu

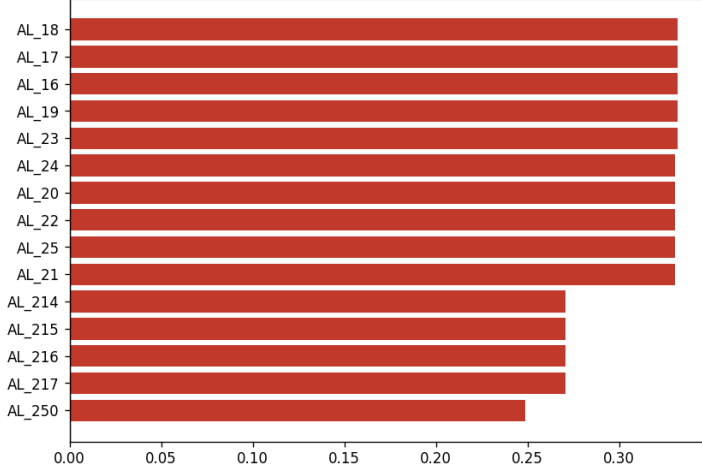


## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

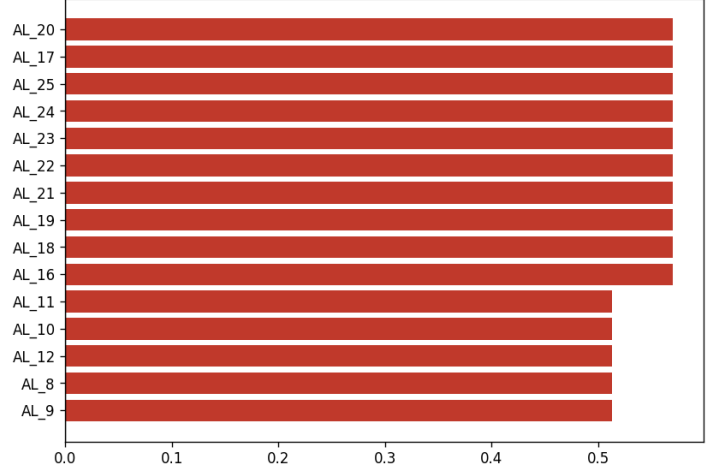
SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### Panel Basina En Guclu 15 Sutun (delta sirali)

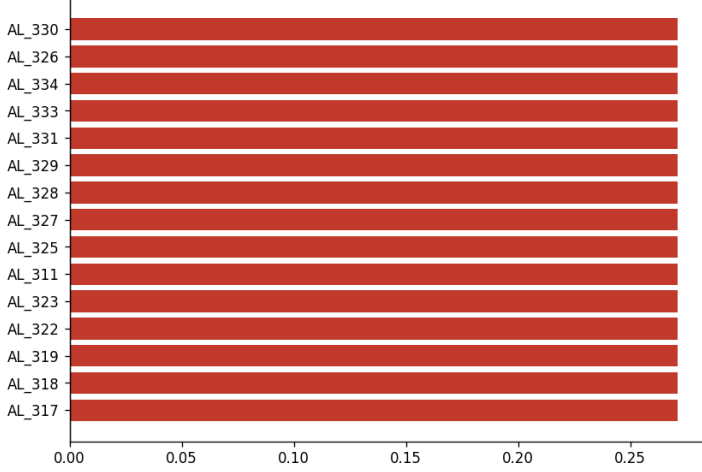
MASTER: En Guclu 15 Sutun (delta = patho-benign eksiklik farki)



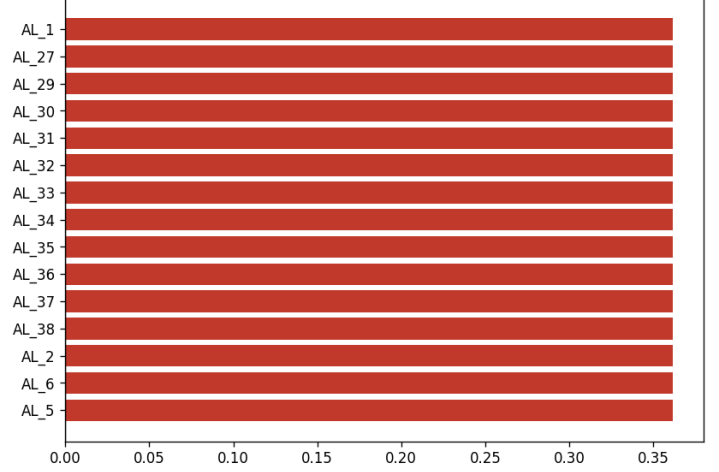
KANSER: En Guclu 15 Sutun (delta = patho-benign eksiklik farki)



PAH: En Guclu 15 Sutun (delta = patho-benign eksiklik farki)



CFTR: En Guclu 15 Sutun (delta = patho-benign eksiklik farki)



## NB43: Missing-Flag Korelasyon Analizi - Tum Paneller

SEED=42 | 4 panel x 2 grup (G\_HIGH >%50 null, G\_LOW <=%50 null) | sabit+ozdes sutun temizligi uygulandi

### 4. Sonuc ve M3 Degerlendirmesi

BULGU: Su panellerde G\_LOW (<=%50 null) grubu da guclu sinyal tasiyor: KANSER (mean\_abs\_phi=0.253).

SONUC: M3'un %50 esigi bu panellerde degerli flag'leri kaciriyor olabilir. Esigi dusurmek veya phi-tabanlı secici flag seti kurmak için yeni bir ablasyon (NB44) önerilir.

Not: KANSER AL\_16..25 blogu (AL\_MISSINGNESS\_LEAKAGE\_RISK) için bu analiz, blogun eksiklik-etiket korelasyonunu nicel olarak teyit/reddeder (Bolum 1'deki saglama noktasına bakınız). CFTR sonuclari n=21 benign nedeniyle dusuk-guven ile yorumlanmalidir.

Bu revizyonda sabit+ozdes sutun temizligi uygulandi (Bolum 0b) -- ozdes AL bloklarinin tekrar eden phi degerleri artik grup ortalamalarini yapay olarak sismemektedir.